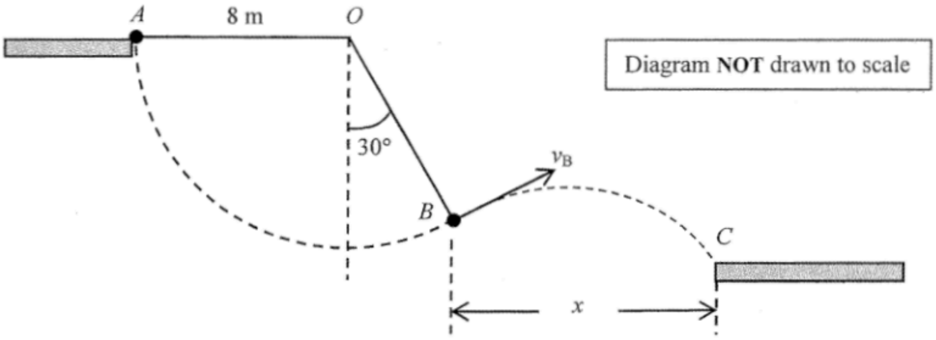
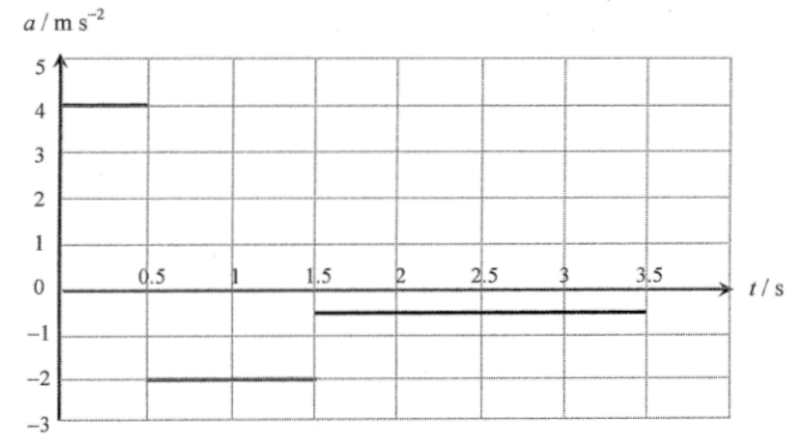
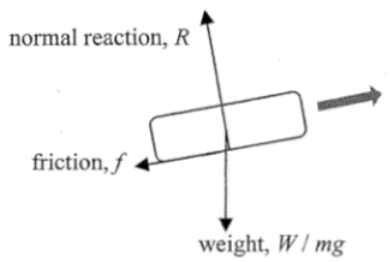
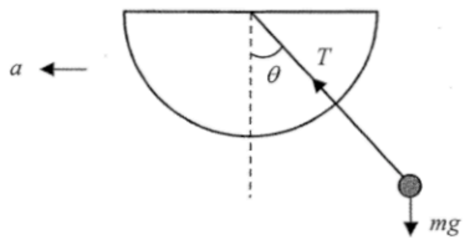
2. (a) 210 atm × (1.0×10^4 cm^3) = 2.0 atm × V	1M	Accept ans. without considering residual volume, i.e. 1.05×10^6 (cm^3)
$V = 1.05 \times 10^6$ cm^3		
Volume available = $1.05 \times 10^6 - 1.0 \times 10^4$		
$= 1.04 \times 10^6$ (cm^3)	1A	
	2	
(b) (i) $V_0 = 1.04 \times 10^6$ cm^3 ÷ 60	1M/1A	V_0 = 17500 (cm^3) if residual volume not considered.
$= 17333 \approx 17300$ (cm^3) (per minute)		
	1	
(ii) V' : total volume of air at this depth/in this situation		$V' = 4.60 \times 10^5$ cm^3 and length of time = 26.3 min. if residual volume not considered.
$\frac{P_1V_1}{T_1} = \frac{P_2V'}{T_2}$		
$\frac{210 \times (1.0 \times 10^4)}{273 + 24} = \frac{4.5 \times V'}{273 + 20}$	1M	
$V' = 4.60 \times 10^5$ cm^3		
Volume available = $4.60 \times 10^5 - 1.0 \times 10^4$		
$= 4.50 \times 10^5$ (cm^3)		
Length of time :		
$= \frac{4.50 \times 10^5}{17333}$	1M	
= 26.0 (min.)	1A	
	3	

3. (a)



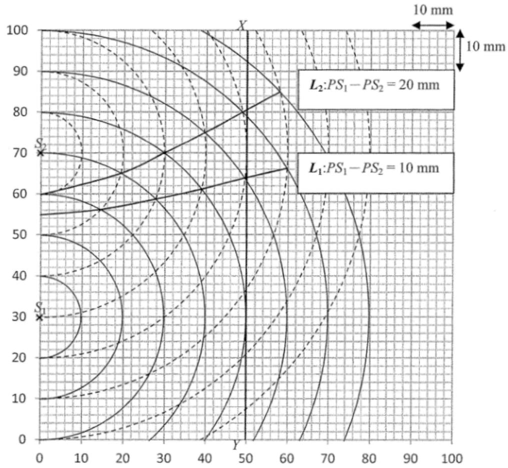
v_B correctly drawn with label (roughly perpendicular to OB).	1A	
$\frac{1}{2}mv^2 = mgh$		
$v_B{}^2 = 2gh = 2 \times 9.81 \times (8\cos30°)$	1M	
$v_B = 11.7\ \text{m s}^{-1}$ (11.65896) (or $11.77\ \text{m s}^{-1}$ for $g = 10\ \text{m s}^{-2}$)	1A	
		3
(b) (i) $x = v_x t = 11.7\cos30° \times 1.25$ [$v_x = v_B\cos30°$]	1M	
$= 12.6$ m (12.62119)	1A	
(or 12.7 to 12.8 m for $g = 10\ \text{m s}^{-2}$)		
		2
(ii) $y = ut - \frac{1}{2}gt^2$		
$u = v_y = v_B\sin30° = 5.83\ \text{m s}^{-1}$	1M	
$y = v_y(1.25) - \frac{1}{2}(9.81)(1.25)^2$	1M	
$y = -0.38$ m (-0.414 to -0.352 m)	1A	
(or -0.455 to -0.4375 m for $g = 10\ \text{m s}^{-2}$)		
Platform C is 0.38 m below B.		3
(c) Total mechanical energy is the same / unchanged.	1A	
		1

			Marks	
4.	(a)	The block decelerates uniformly (until at rest or its velocity is zero at $t = 1.5$ s).	1A	
		It then moves with uniform acceleration down the plane (until $t = 3.5$ s).	1A	
			2	
	(b)	(i) $a_2 = \left\lvert \dfrac{-1-0}{3.5-1.5} \right\rvert$	1M	
		$= 0.5 \text{ m s}^{-2}$	1A	
			2	
		(ii) 		
		Either acceleration correct	1A	
		All correct	1A	
			2	
	(c)			
		Friction correctly indicated	1A	
		All correct	1A	
			2	
	(d)	Moving upward : $-mg \sin\theta - f = ma$	1M	
		$-(1)(9.81) \sin\theta - f = (1)(-2)$……………… ①		
		Moving downward : $-mg \sin\theta + f = ma'$	1M	
		$-(1)(9.81) \sin\theta + f = (1)(-0.5)$……………… ②		
		② − ① : $2f = 1.5$		
		$f = 0.75$ N	1A	
		(Note: $\theta = 7.32°$)		
			3	

5. Diagram		
	1A	
Tie one end of the string to the metal ball and the other end through the centre/hole of the protractor.	1A	
When the train is at rest, held fixed the protractor in the plane along the direction of motion such that the string is on, say, the 90° mark.	1A	
When the train is accelerating with acceleration a, the string will make an angle, say θ, with the vertical. Measure the angle θ.	1A	
Let T be the tension of the string, Vertically : $T\cos\theta = mg$ .... ① Horizontally : $T\sin\theta = ma$.............. ② where m is the mass of the ball	1M	
$\frac{②}{①}$: $\tan\theta = \frac{a}{g}$ $a = g\tan\theta$	1A	
	6	

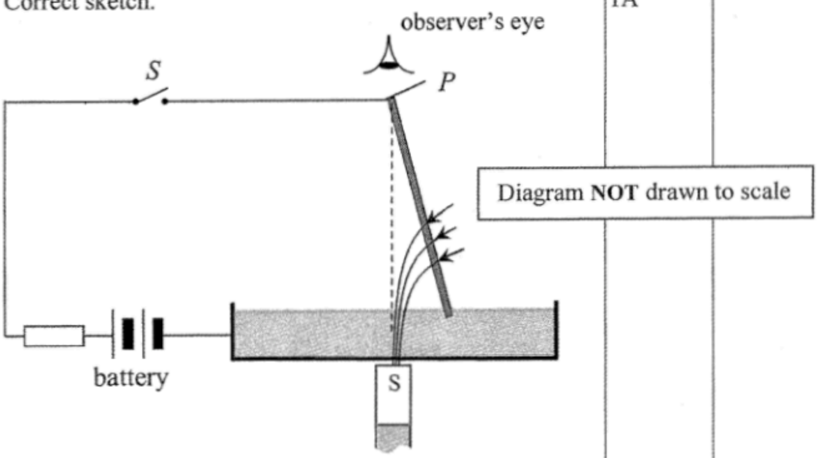
6. (a) - high temperature gradient; <u>or</u> - long path lengths for light rays.	1A		
<u>Or</u> total internal reflection occurs.	1A		
		1	
(b) (i) $n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3 = n_4 \sin \theta_4$	1M		
$\sin \theta_1 = \frac{n_4}{n_1} \sin \theta_4$			
$\theta_1 = \sin^{-1} (\frac{1.000221}{1.000261})$	1M		
$= 89.5° \ (89.488°)$	1A		
		3	
(ii) $\frac{h}{L} = \tan \alpha = \frac{1}{\tan \theta_1}$	1M		
$L = h \tan \theta_1 = 1.5 \tan 89.5° = 167.72$			
$= 168 \text{ m}$	1A		Accept 167.7 m to 172.0 m
<u>Or</u> $L = \frac{h}{\tan \alpha} = \frac{1.5}{\tan 0.5°} = 171.88$	1M		
$= 172 \text{ m}$	1A		
		2	
(c) The same distance away (168 m) because	1A		
the illusion of 'water source' is caused by reflection of the light of distant objects at the same fixed angle. [i.e. $\alpha = 90° - 89.488° = 0.512°$ with the horizontal]	1A		
<u>Or</u> As long as the conditions for bending of the light and internal reflection are still the same,	1A		
the 'water source' remains 168 m away (satisfying the same conditions / angle of reflection).	1A		
		2	

	Marks	Remarks
7. (a) $\Delta y = \frac{\lambda D}{a}$	1M	
$= \frac{(650\times10^{-9})\times3.0}{0.325\times10^{-3}}$		
$= 0.006$ m or 6 mm	1A	
	2	
(b) The screen is uniformly illuminated (The interference patterns exist very briefly and change rapidly such that, to human eyes, they are averaged out).	1A	NOT accept 'interference does not occur'.
The light from the LEDs is incoherent (i.e. no fixed phase relationship between the light coming out from the two LEDs).	1A	
	2	
(c) path difference $PS_1 - PS_2 = 10$ mm, L_1 correct.	1A	
path difference $PS_1 - PS_2 = 20$ mm, L_2 correct.	1A	
Constructive interference (occurs at P)	1A	
	3	



	Marks
(d) (i) $\Delta y = y_2 - y_1 = 31\text{ mm} - 14\text{ mm} = 17\text{ mm} \pm 2$ mm	1A
	1
(ii) Screen has to be far away from slits or $D >> a$, (i.e. to satisfy $D >> y$ / consider y to be close to the central maximum)	1A
Or screen is too close to slits or $D >> a$ not satisfied (i.e. $D >> y$ not satisfied)	1A
Make use of small angle approximation ($\theta \approx$) $\sin\theta \approx \tan\theta$ / small angle approximation cannot be applied.	1A
	2

	Marks
8. (a) (i) $\rho = \frac{RA}{l}$	
$\frac{R}{l} = \frac{\rho}{A} = \frac{2.6\times10^{-8}}{1.3\times10^{-5}}$	
$= 2.0 \times 10^{-3}\ \Omega\ \text{m}^{-1}$	
$= 2.0\ \Omega\ \text{km}^{-1}$ or $2.0\ \Omega$	1A
	2
(ii) The strands of transmission lines are in parallel / The cross-sectional area of cable is larger than that of each transmission line / Resistance is inversely proportional to the cross-sectional area of the cable	1A
$R_{\text{cable}} = \frac{R}{40} = 0.05\ \Omega\ \text{km}^{-1}$ or $0.05\ \Omega$	1M
$(\frac{1}{R_{\text{cable}}} = \frac{1}{R} + \frac{1}{R} + \ldots + \frac{1}{R} \rightarrow \frac{1}{R_{\text{cable}}} = \frac{40}{R})$	
	2
(iii) The resistance of the bird's body is much larger than that of a short segment of the overhead cable.	1A
Or The bird is in parallel with a short segment of the overhead cable. The potential difference across its feet is very small (very small resistance per km).	1A
Hence, negligible current flows through the bird's body.	1A
	2
(b) (i) $I = \frac{P}{V} = \frac{180\times10^6}{400\times10^3}$	1M
$= 450$ A	1A
	2
(ii) Percentage of power loss $= \frac{P_{loss}}{P_{total}}\times100\%$	1M
$= \frac{450^2\times0.05\times10}{180\times10^6}\times100\%$	
$= 0.05625\ \% < 0.1\ \%$	1A
	2
(iii) (I) $N_p : N_s = V_p : V_s$ $12 : 1 = 400 : V_s$ $V_s = 33.3$ kV	1A
	1
(II) Any **ONE** of the followings: Resistance of coils + use thicker wire for the coils / Magnetisation and demagnetisation of core + use soft iron core / Induced eddy currents in core + laminated core / Flux leakage + core design	1A+1A
	2

9.	(a)	Right (current flowing downward, B-field into paper) When the rod reaches the highest point it falls. Then its lower end touches the conducting liquid again and the same magnetic force makes it 'kick' out from the liquid. The process repeats so the rod continually 'kicks' out and then returns.	1A 1A 1A		
				3	
	(b) (i)	As moment $= F \times d$ 7.2×10^{-4} N m $= F\,(0.09\text{ m})$ $F = \dfrac{7.2\times10^{-4}}{0.09} = 8.0 \times 10^{-3}$ N	 1M 1A		
				2	
	(ii)	$F = BIl$ $8.0 \times 10^{-3}\text{ N} = B\,(3.2\text{ A})\,(0.06\text{ m})$ $B = 0.042$ T	1M 1A		
				2	
	(c) (i)	Correct sketch.  	1A		
				1	
	(ii)	The rod will rotate anti-clockwise (as viewed from above).	1A		
		$\boxed{\text{Or The rod describes a conical pendulum.}}$	1A		
				1	

10. (a)	Mass deficit $= (2.014102 + 3.016049)\text{ u} - (4.002602 + 1.008665)\text{ u}$ $= 0.018884\text{ u}$	1M	
	Energy released $= 0.018884 \times 931$ MeV $= 17.58$ (MeV)	1A	
	Or Energy released $= 0.018884 \times 1.661 \times 10^{-27} \times c^2$ $= 2.823 \times 10^{-12}$ J or 17.64 MeV	1A	
		2	
(b) (i)	To overcome the (electrostatic) repulsion between the two (positive) nuclei and	1A	
	becomes electrical potential energy (of the two nuclei).	1A	
		2	
(ii)	High temperature enables them to have sufficient kinetic energy (to overcome electrical repulsion between the nuclei).	1A	
		1	
(iii)	Kinetic energy becomes electrical potential energy $E_p = 2 \times \frac{1}{2} m (c_{rms})^2 = 2 \times \frac{3RT}{2N_A}$	1M	Accept without the factor '×2'
	$0.4\text{ MeV} = 2 \times (\frac{3 \times 8.31 \times T}{2 \times 6.02 \times 10^{23}})$		0.4 MeV = 6.4×10^{-14} J
	$T = 1.545 \times 10^9$ K i.e. order of magnitude 10^9 (K)	1A	
	Alternatively: $E_p = \frac{1}{4\pi\varepsilon_0} \cdot \frac{e^2}{10^{-15}} = 2 \times \frac{3RT}{2N_A}$	1M	
	$T = 5.56 \times 10^9$ K i.e. order of magnitude 10^9 (K)	1A	
		2	